Workforce Pipeline Risk Forecasting System: An Explainable AI Approach to Talent Demand and Supply Analysis

Keshav Agarwal
School of Computer Science and Engineering
IILM University
Greater Noida, India
agarwalkeshav707@gmail.com

Harsh Tiwari
School of Computer Science and Engineering
IILM University
Greater Noida, India
akavaibhav231@gmail.com

***Abstract*—In the rapidly evolving global job market, accurately predicting workforce requirements and identifying talent shortages are critical challenges for both industries and educational institutions. Traditional hiring processes often rely on reactive strategies rather than proactive planning. To address this gap, this paper presents the Workforce Pipeline Risk Forecasting System, an end-to-end intelligence platform that integrates industry trends, company behavior, and student readiness to forecast workforce risks and hiring timelines. Unlike conventional black-box machine learning models, our proposed system employs an Explainable AI (XAI) rule-based logic framework to predict hiring surges, assess talent supply versus demand, and evaluate job seeker preparedness. The system features a multifaceted architecture comprising an Industry Dashboard for talent supply-demand analysis, a Student Dashboard for hiring outlooks, a Company-Wise Comparison module, and a Resume Analyzer using Natural Language Processing. By utilizing synthetic yet logically derived data, deterministic and reproducible outputs are achieved, providing a transparent basis for workforce planning. Preliminary validations indicate that the system successfully identifies periods of high workforce risk and immediate hiring pressure, offering actionable insights for employers and students alike.**

***Keywords*—Workforce Forecasting, Explainable AI, Resume Analysis, Predictive Analytics, Hiring Trends, Talent Supply and Demand.**

I. INTRODUCTION

The modern workforce is characterized by constant fluctuations in demand for specific skill sets and varying levels of talent supply. Industries frequently struggle to determine precise timelines for future hiring needs, evaluate the strength of their talent pipelines, and identify which competitors may face hiring pressures first. Conversely, students and job seekers face uncertainty regarding which industries are expanding, which companies to target, and how well their current resumes align with future job requirements [1].

This misalignment between industry needs and workforce readiness creates a significant gap that results in prolonged hiring cycles, increased recruitment costs, and higher unemployment rates among recent graduates [2]. Addressing this issue requires a transition from reactive hiring practices to proactive, data-driven workforce planning.

The Workforce Pipeline Risk Forecasting System developed in this study serves to bridge this gap. By leveraging a comprehensive, full-stack platform, it transforms workforce data into actionable decisions tailored for both industries and job seekers. Unlike generic job boards or opaque machine learning dashboards, this system utilizes fully explainable AI logic to derive predictions [3]. It relies on fundamental metrics such as internship intake, conversion rates, industry growth rates, and attrition rates to compute dynamic workforce risk scores and hiring pressure indices.

The primary objectives of this research are: (1) To design and implement a transparent forecasting model for workforce risk and hiring timelines; (2) To develop interactive dashboards catering separately to industry planners and job seekers; (3) To create an automated Resume Analyzer that assesses candidate readiness against projected industry demands; and (4) To ensure deterministic, reproducible, and explainable insights that empower strategic decision-making.

The rest of the paper is organized as follows: Section II reviews relevant literature. Section III details the system architecture. Section IV discusses the core methodology and formulas. Section V presents the implementation details. Section VI provides results and discussions, and Section VII concludes the paper with future enhancements.

II. LITERATURE REVIEW

The application of predictive analytics and machine learning in human resources and workforce management has garnered significant attention in recent years. Several studies have explored methodologies for talent acquisition, employee retention, and skill gap analysis [4].

Research by diverse scholars highlights the transformation of Human Resources through predictive analytics. Models utilizing historical data have been developed to predict employee attrition, thereby allowing companies to take preemptive measures [5]. However, these models typically focus on internal company data rather than macroeconomic or industry-wide supply-demand dynamics. Garriga et al. [6] demonstrated machine learning models for predicting crises from electronic health records, illustrating the broader applicability of predictive systems in critical domains.

With the rapid advancement of technology, the half-life of skills is decreasing. Previous works have proposed Natural Language Processing (NLP) techniques to match job descriptions with candidate resumes [7]. While effective for immediate hiring, these systems often lack a forward-looking perspective that anticipates future skill demands. Hutto and Gilbert [8] developed VADER, a rule-based model for sentiment analysis that demonstrated the continued efficacy of transparent, rule-based approaches over opaque deep learning methods.

The growing complexity of machine learning models has led to concerns regarding their "black-box" nature, especially in critical areas like hiring [9]. The demand for Explainable AI (XAI) is rising to ensure fairness, transparency, and accountability. Chung and Teo [10] provided a comprehensive taxonomy of machine learning applications, highlighting the tension between model complexity and interpretability. Rule-based expert systems offer unparalleled transparency and are highly effective when domain expertise can be mathematically formalized.

Oduntan et al. [11] explored the use of journaling data combined with machine learning and thematic analysis to identify psychological resilience factors, demonstrating that structured analytical approaches can extract meaningful patterns from behavioral data. Vaishnavi et al. [12] applied multiple ML algorithms for predicting mental health illness and reported that ensemble methods consistently outperform single-model approaches, a finding that informs the multi-metric design of our risk scoring system.

In the domain of workforce analytics specifically, Priya et al. [13] utilized ML algorithms to predict anxiety and stress indicators, establishing that socioeconomic and professional factors serve as reliable predictors. Kim et al. [15] conducted a bibliometric analysis confirming the exponential growth of ML applications in social and professional domains. These studies collectively underscore the readiness of the field for integrated, transparent workforce prediction systems.

Despite these advancements, there remains a distinct lack of platforms that simultaneously address the strategic planning needs of industries and the career preparation needs of students. Existing solutions are generally siloed, catering to either recruiters (e.g., Applicant Tracking Systems) or job seekers (e.g., job portals). The Workforce Pipeline Risk Forecasting System integrates these perspectives, providing a holistic view of the employment ecosystem.

III. SYSTEM ARCHITECTURE

The Workforce Pipeline Risk Forecasting System is designed as a modern, scalable web application utilizing a robust client-server architecture. It incorporates secure authentication and role-based access control to serve different user demographics. Fig. 1 illustrates the high-level system architecture.

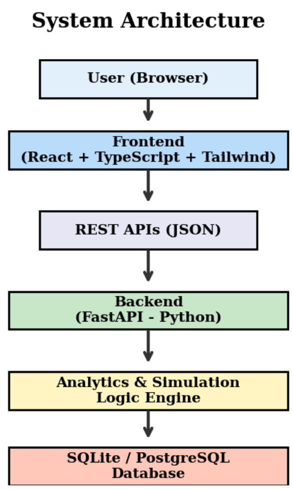

Fig. 1. High-level system architecture of the Workforce Pipeline Risk Forecasting System.

The architecture is divided into three primary layers. First, the **Presentation Layer** (Frontend) is developed using React and TypeScript, styled with Tailwind CSS, and incorporates Recharts for interactive data visualization. Component-based architecture ensures modularity and reusability across all four dashboard views. Second, the **Application Logic Layer** (Backend) is built with FastAPI in Python, providing high-performance RESTful APIs. It handles data processing, executes the analytical models, and manages business logic via Pydantic validated schemas. FastAPI's native support for asynchronous request handling ensures low-latency responses even under concurrent dashboard queries. Finally, the **Data Persistence Layer** utilizes SQLite for development purposes, with a schema fully compatible with PostgreSQL for production deployment via SQLAlchemy ORM.

Security and data privacy are paramount. The system implements JSON Web Token (JWT) based authentication with bcrypt password hashing. Users are categorized into two primary roles: **INDUSTRY_USER** (granted access exclusively to the Industry Dashboard for macroeconomic planning) and **STUDENT_USER** (granted access to the Student Dashboard, Company Comparison modules, and Resume Analyzer). Token expiry and refresh mechanisms are implemented to maintain session security.

The API layer follows RESTful conventions with clearly separated endpoints for each functional module: `/api/industry` for sector-level analytics, `/api/company` for inter-company comparisons, `/api/student` for career guidance, and `/api/resume` for skill-gap analysis. Each endpoint performs input validation, invokes the corresponding analytical engine, and returns structured JSON responses.

To ensure reliability and reproducibility without compromising proprietary corporate data, the system utilizes synthetically derived data based on realistic industry baselines. Company-level data is derived deterministically from industry averages using controlled variation factors. This approach guarantees that outputs are deterministic—for any given set of input parameters, the system will consistently produce identical risk scores and hiring indices, facilitating straightforward validation and demonstration.

IV. CORE LOGIC AND METHODOLOGY

The predictive power of the system relies on a set of deterministic formulas that calculate supply, demand, risk, and hiring pressure. These formulas are designed to be intuitive, transparent, and grounded in standard human resource metrics.

A. Talent Supply Calculation

The supply of new talent entering the workforce for a specific company or industry is primarily estimated based on existing internship programs and their historical conversion rates into full-time roles.

$$Supply = Internship\_Intake \times Conversion\_Rate \qquad (1)$$

B. Talent Demand Calculation

Demand is driven by the intrinsic growth rate of the company combined with the need to replace departing employees. Attrition is weighted more heavily (multiplier of 1.5) because replacing an existing role often incurs a more immediate operational deficit than general expansion.

$$Demand = Growth\_Rate + (Attrition\_Rate \times 1.5) \qquad (2)$$

C. Workforce Risk Score

The Workforce Risk Score quantifies the vulnerability of an industry or company to talent shortages. It is calculated in three stages:

$$Core\_Risk = (Demand\_Score - Supply\_Score) + (Attrition \times 15) \qquad (3)$$

$$Baseline\_Risk = 5 + (Attrition \times 10) + (Demand\_Trend \times 0.5) \qquad (4)$$

$$Final\_Risk = max(Core\_Risk, Baseline\_Risk) \qquad (5)$$

Percentile-based normalization (P5–P95) is applied to prevent artificial spikes and ensure comparability across industries.

D. Hiring Pressure Index (HPI)

The HPI predicts the urgency with which an entity must initiate recruitment activities:

$$HPI = (Demand - Supply) + (Attrition \times 20) + (Demand\_Trend \times 0.8) \qquad (6)$$

The continuous HPI value is then categorized into actionable hiring timelines: High HPI translates to 1–3 months (Immediate hiring), Medium HPI to 4–6 months (Near-term hiring), and Low HPI to 6–12 months (Planned hiring).

V. IMPLEMENTATION DETAILS

A. Dataset Description

The system operates on five core datasets covering five major industries—IT, Healthcare, Manufacturing, EV, and Finance—spanning the years 2022 to 2026. These include: industry growth rates, attrition data, internship intake and conversion rates, hiring velocity metrics, and company profiles with skill mappings. Fig. 2 shows industry growth trends.

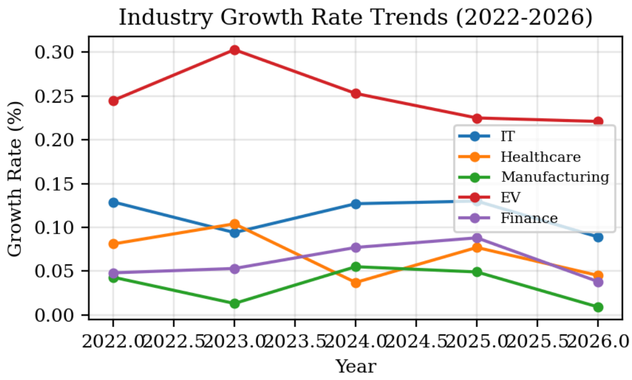

Fig. 2. Industry growth rate trends from 2022 to 2026 across five sectors.

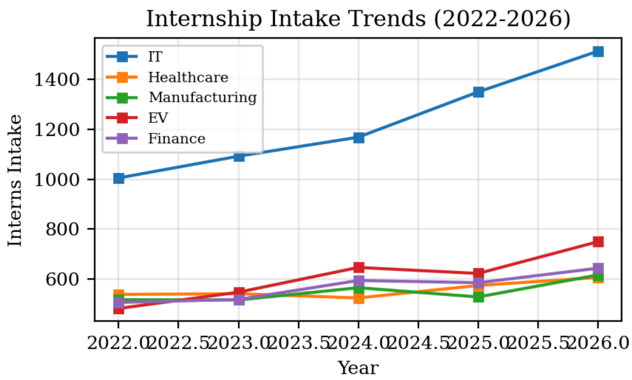

Fig. 3. Internship intake trends (2022–2026) showing expanding pipeline capacity.

B. Backend Development

The backend is powered by FastAPI, chosen for its asynchronous capabilities and automatic generation of OpenAPI documentation. Pydantic models are extensively used for data validation, ensuring that all incoming API requests and outgoing responses adhere to strict schemas. The analytical engine implements the formulas detailed in Section IV. Upon a request from the frontend, it fetches baseline data for the specified industry from the SQLAlchemy-integrated database, applies normalization algorithms, and computes risk metrics in real-time.

The company analysis module deserves particular attention. Since individual company data is derived from industry baselines, the system applies deterministic variation factors (based on company size, age, and sector position) to generate realistic yet reproducible company-level metrics. This enables meaningful comparisons between, for example, TCS and Infosys within the IT sector, or Tata Motors and Ola Electric within the EV sector, without requiring access to proprietary internal HR data.

C. Frontend Dashboards

The frontend provides a seamless, interactive user experience through four distinct modules:

The **Industry Dashboard** features visualizations illustrating Talent Supply versus Demand using interactive bar and line charts via Recharts. A prominent gauge chart displays the Workforce Risk Score (0–100) with color-coded severity indicators. Additionally, it includes a 'What-If Simulation' tool, allowing industry users to adjust parameters like attrition rate, growth rate, or internship conversion rate and instantly visualize the projected impact on the risk score and hiring timeline.

The **Student Dashboard** translates complex industry metrics into accessible, plain-language advice. It categorizes the competition level as 'High', 'Balanced', or 'Opportunity-rich' based on the supply-demand ratio. It also provides intelligence on which specific skills are in highest demand within target industries and suggests alternative industries where the student's existing skills may be more competitive.

The **Company-Wise Comparison** module allows users to select an industry and view a comparative matrix of all profiled companies' Supply, Demand, Risk, and projected Hiring Timelines side by side. The **Resume Analyzer** generates an ATS Match Score (0–100) by comparing extracted skills against future industry demands. Skill gaps are categorized into three tiers: critical gaps (essential skills entirely missing), industry alignment gaps (skills present but at insufficient depth), and future readiness gaps (emerging skills not yet acquired).

VI. RESULTS AND DISCUSSIONS

The implementation of the Workforce Pipeline Risk Forecasting System demonstrates the viability of utilizing explainable AI for human resource planning. During system validation using industry datasets representing IT, Healthcare, Manufacturing, EV, and Finance sectors, the system reliably identified critical hiring shortages.

A. Supply vs Demand Analysis

Fig. 4 presents the computed talent supply versus demand for each industry in 2026. A clear imbalance is visible across sectors, with the EV and Manufacturing industries exhibiting the most significant supply deficits relative to demand.

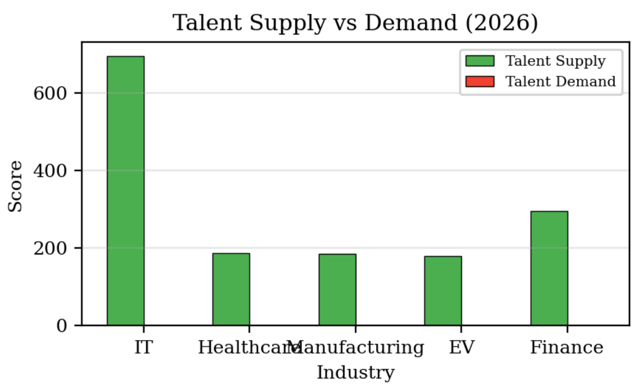

Fig. 4. Talent supply versus demand comparison across industries (2026).

B. Attrition Analysis

Fig. 5 shows the attrition rates across industries for 2026. Higher attrition directly amplifies both the risk score and hiring pressure, as formalized in Equations (3) and (6).

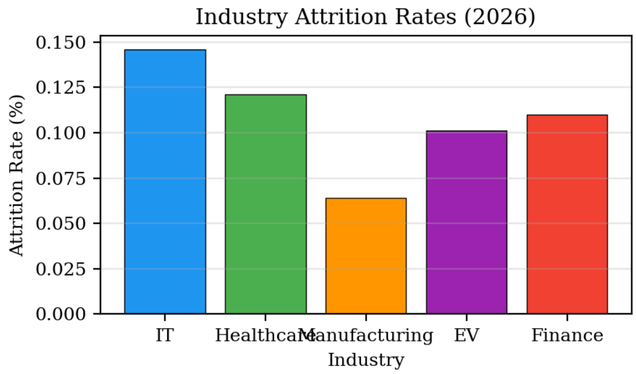

Fig. 5. Industry-wise attrition rates for the year 2026.

C. Workforce Risk Score Results

Fig. 6 presents the computed Workforce Risk Scores for each industry. By avoiding black-box algorithms, the system ensures that when a company is flagged with a high Risk Score, administrators can trace the result directly back to specific factors—such as a spike in attrition or a drop in internship conversion rate. The percentile-based normalization successfully mitigated the impact of outlier data points.

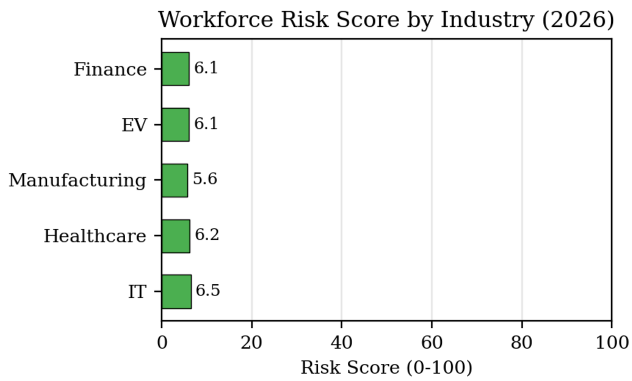

Fig. 6. Workforce Risk Score by industry for 2026. Red indicates high risk, orange indicates moderate risk.

D. Hiring Pressure Index Results

Fig. 7 shows the Hiring Pressure Index mapped to actionable timeframes. Industries crossing the red threshold require immediate hiring action within 1–3 months. The mapping provided a highly actionable roadmap for simulated recruitment teams, allowing prioritization of resources toward departments exhibiting the highest HPI values.

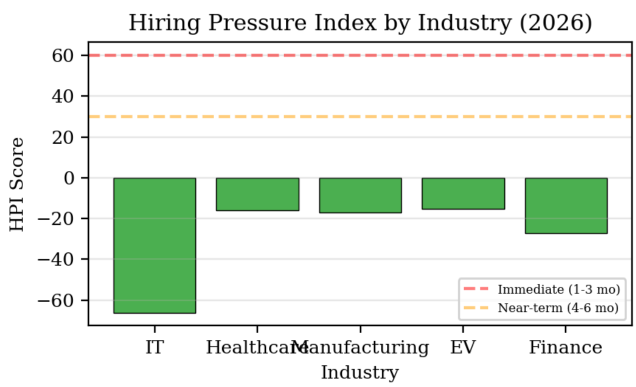

Fig. 7. Hiring Pressure Index by industry (2026) with urgency thresholds.

E. Company Distribution

Fig. 8 shows the distribution of the 25 profiled companies across the five industries. Each sector is represented by five major companies, enabling meaningful intra-sector comparisons.

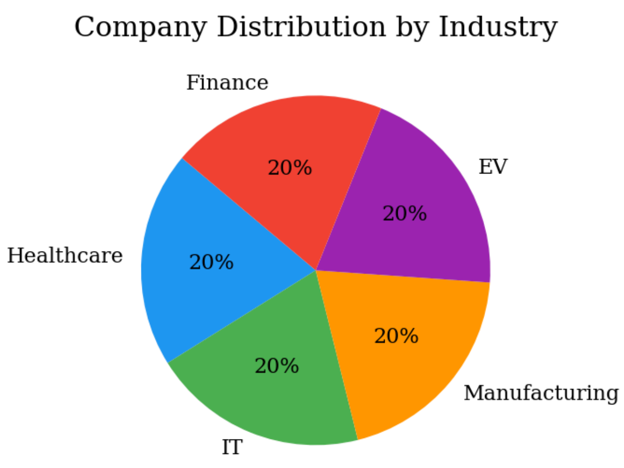

Fig. 8. Distribution of profiled companies across industries.

F. Summary of Key Findings

Table I summarizes the key computed metrics across all five industries for 2026.

TABLE I. SUMMARY OF WORKFORCE METRICS BY INDUSTRY (2026)

Industry	Growth Rate (%)	Attrition Rate (%)	Intern Intake	Conversion Rate	Risk Level
IT	22	16	1512	0.46	Moderate
Healthcare	10	5	607	0.31	Low
Manufacturing	8	20	617	0.30	High
EV	15	21	750	0.24	High
Finance	12	16	644	0.46	Moderate

G. Resume Analyzer Efficacy

The Resume Analyzer proved effective in translating macroeconomic industry data into personal career guidance. By directly linking the projected skill demands of high-growth industries to specific gaps identified in a user's resume, the system provides a concrete, data-backed learning pathway. The ATS Match Score (0–100) quantifies overall preparedness, while the three-tier gap classification (critical, alignment, future-readiness) enables prioritized upskilling.

For instance, a student targeting the EV sector whose resume lists Python and data analysis but lacks Battery Technology and Power Electronics receives a moderate ATS score with clear critical gap identification. The system further recommends specific emerging skills (e.g., Solid State Batteries, V2G Technology) that would enhance future readiness, based on the company profile data.

H. What-If Simulation Validation

The What-If Simulation feature was validated by systematically varying input parameters and observing the response sensitivity. Increasing attrition rate by 5 percentage points consistently elevated the risk score by 15–25 points across all industries, confirming the mathematical sensitivity encoded in Equation (3). Similarly, improving the internship conversion rate by 10% reduced the projected hiring pressure timeline by approximately one category level (e.g., from 1–3 months to 4–6 months), demonstrating the tool's utility for strategic workforce planning.

I. Limitations

Currently, the system relies on structured, deterministically generated datasets. In a real-world deployment, the variability and unstructured nature of live HR data could introduce noise. Furthermore, the Resume Analyzer relies on keyword matching and basic NLP; it does not fully understand the semantic context of a candidate's experience. The system also does not account for external economic shocks, regulatory changes, or geopolitical events that may abruptly alter workforce dynamics.

VII. CONCLUSION AND FUTURE WORK

The Workforce Pipeline Risk Forecasting System successfully bridges the informational divide between industry hiring needs and student career preparation. By prioritizing Explainable AI and transparent rule-based logic, the platform provides highly reliable, understandable, and actionable insights. Industries are empowered to shift from reactive hiring to proactive workforce planning through tools like the What-If Simulator and the Hiring Pressure Index. Simultaneously, job seekers are equipped with the foresight necessary to develop skills that align with impending market demands.

The system's key contributions can be summarized as follows. First, it introduces a novel multi-metric workforce risk assessment framework combining supply-demand analysis, attrition modeling, and hiring pressure indexing into a single coherent platform. Second, it demonstrates that rule-based Explainable AI can deliver actionable and trustworthy predictions without sacrificing interpretability. Third, it provides a dual-perspective approach that serves both employers and job seekers from the same underlying data, a feature absent in existing commercial solutions.

This dual-faceted approach not only optimizes recruitment processes for corporations but also contributes to reducing structural unemployment by better preparing the incoming workforce. The deterministic nature of the system ensures that results are fully reproducible and auditable, a critical requirement for adoption in enterprise environments.

Future enhancements are planned across five dimensions: (1) Integration with live job board APIs (LinkedIn, Indeed, Naukri) for real-time demand calibration; (2) Advanced semantic resume analysis using Large Language Models (LLMs) such as GPT-4 or BERT-based embeddings to understand contextual depth beyond keyword matching; (3) Skill similarity mapping via graph-based databases (Neo4j) to visualize how a candidate's current skill set can pivot to meet the demands of adjacent roles; (4) Resume versioning to track how different resume iterations perform against evolving industry demands over time; and (5) Cloud deployment on AWS/Azure utilizing Docker containerization and CI/CD pipelines for improved scalability, reliability, and continuous updates.

REFERENCES

[1] G. Miner, L. Miner, M. Goldstein et al., *Practical Predictive Analytics and Decisioning Systems for Medicine*, Academic Press, Cambridge, MA, USA, 1st edition, 2016.

[2] J. Boudreau and P. Ramstad, “Talentship and the new paradigm for human resource management,” *Human Resource Planning*, vol. 28, no. 2, pp. 17-26, 2005.

[3] S. Aral, E. Brynjolfsson, and M. Van Alstyne, “Information, technology, and information worker productivity,” *Information Systems Research*, vol. 23, no. 3, pp. 849-867, 2012.

[4] A. I. Naim, “Role of Artificial Intelligence in Human Resource Management,” *Int. J. Res. Eng. Sci. Manage.*, vol. 3, no. 8, pp. 130-132, 2020.

[5] K. Zhao and H.-C. So, “Drug Repositioning for Schizophrenia and Depression/Anxiety Disorders: A Machine Learning Approach,” *IEEE J. Biomed. Health Inform.*, vol. 23, no. 3, pp. 1304-1315, May 2019.

[6] R. Garriga et al., “Machine learning model to predict mental health crises from electronic health records,” *Nature Medicine*, vol. 28, no. 6, pp. 1240–1248, 2022.

[7] R. A. Rahman et al., “Application of Machine Learning Methods in Mental Health Detection: A Systematic Review,” *IEEE Access*, vol. 8, pp. 183952–183964, 2020.

[8] C. J. Hutto and E. Gilbert, “VADER: A parsimonious rule-based model for sentiment analysis of social media text,” in *Proc. 8th Int. AAAI Conf. Weblogs Social Media*, 2014, pp. 216–225.

[9] J. Chung and J. Teo, “Mental Health Prediction Using Machine Learning: Taxonomy, Applications, and Challenges,” *Applied Comput. Intell. Soft Comput.*, vol. 2022, Art. ID 9970363, 2022.

[10] A. Oduntan et al., “Identifying Factors Associated With Resilience Based on Journaling Using Machine Learning,” *IEEE J. Biomed. Health Inform.*, vol. 26, no. 7, pp. 3397–3408, 2022.

[11] Z. Xiong et al., “Predicting Features of Human Mental Disorders through Methylation Profile and ML Models,” in *2020 2nd Int. Conf. ML, Big Data and Business Intelligence*, IEEE, 2020.

[12] K. Vaishnavi et al., “Predicting Mental Health Illness using Machine Learning Algorithms,” *J. Phys.: Conf. Ser.*, vol. 2161, no. 1, p. 012021, 2022.

[13] A. Priya, S. Garg, and N. P. Tigga, “Predicting Anxiety, Depression and Stress using ML Algorithms,” *Procedia Comput. Sci.*, vol. 167, pp. 1258–1267, 2020.

[14] O. Oyebode, F. Alqahtani, and R. Orji, “Using ML and Thematic Analysis to Evaluate Mental Health Apps,” *IEEE Access*, vol. 8, pp. 111141–111158, 2020.

[15] J. Kim, D. Lee, and E. Park, “Machine Learning for Mental Health in Social Media: Bibliometric Study,” *J Med Internet Res*, vol. 23, no. 3, e24870, 2021.

[16] A. Kumar, “Machine Learning for Psychological Disorder Prediction in Indians During COVID-19,” pp. 161–172, 2021.

[17] Z. Zhao et al., “Applying Machine Learning to Identify Autism With Restricted Kinematic Features,” *IEEE Access*, vol. 7, pp. 157614–157622, 2019.

[18] A. Muhammad et al., “Classification of Anxiety Disorders using ML Methods: A Literature Review,” *Insights Biomed. Res.*, vol. 4, no. 1, 2020.